

MATH0054 Analytical Dynamics

<i>Year:</i>	2025–2026
<i>Code:</i>	MATH0054
<i>Level:</i>	6 (UG)
<i>Normal student group(s):</i>	UG: Year 3 Mathematics degrees PG: (2026-27) MSc Mathematics (level 6)
<i>Value:</i>	15 credits (= 7.5 ECTS credits)
<i>Term:</i>	2
<i>Assessment:</i>	90% examination, 10% coursework
<i>Normal Pre-requisites:</i>	MATH0009, MATH0011
<i>Lecturer:</i>	Prof A Sokal

Course Description and Objectives

Analytical dynamics develops Newtonian mechanics to the stage where powerful mathematical techniques can be used to determine the behaviour of many physical systems. The mathematical framework also plays a role in the formulation of modern quantum and relativity theories. Topics studied are the kinematics of frames of reference (including rotating frames), dynamics of systems of particles, Lagrangian and Hamiltonian dynamics and rigid body dynamics. The emphasis is both on the formal development of the theory and also use of theory in solving actual physical problems.

Recommended Texts

Relevant books are: (i) Gregory, *Classical Mechanics*; (ii) T L Chow, *Classical Mechanics*; (iii) Goldstein, *Classical Mechanics*; (iv) Taylor, *Classical Mechanics*; (v) Marion, *Classical Dynamics of Particles and Systems*

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